

ACTL3182 Wk3

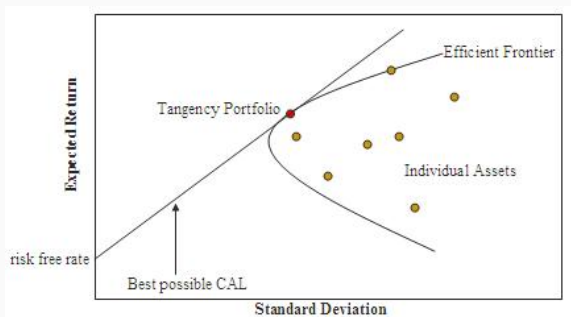
Tadhg XG

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One-Fund Theorem

Recall the one-fund theorem - everyone's investment can be described by a linear combination of the tangency portfolio and the risk free asset.

We can refer to the tangency portfolio as the market portfolio, as everyone's purchases of this single portfolio will sum to the entire market.



Capital Market Line

Capital Market Line

The Capital Market Line is the straight line in (σ, μ) space that passes through the risk-free asset and the market portfolio, that is, if (μ_p, σ_p) is the mean and sd of an efficient portfolio then

$$r_p = r_f + \frac{r_M - r_f}{\sigma_M} \sigma_p.$$

Market Price of Risk

The gradient

$$\frac{r_M - r_f}{\sigma_M}$$

is the market price of risk.

CAPM

After assuming a lot of things, we have for some asset i

$$r_i - r_f = \beta(r_M - r_f),$$

where

$$\beta = \frac{\sigma_{iM}}{\sigma_M^2} = \frac{\text{Cov}(\tilde{r}_M, \tilde{r}_i)}{\text{Var}(\tilde{r}_M)}$$

CAPM can be used to derive the current price of the asset (shown in lectures and later).

Factor Model

Single Factor Model

Let f be the random return of some factor. Then, for some asset i , we assume

$$\tilde{r}_i = \alpha_i + \beta_i f + \epsilon_i,$$

where α_i, β_i are asset specific constants and ϵ_i is a random variable with the following:

- Zero mean: $\mathbb{E}[\epsilon_i] = 0$;
- Constant variance: $\text{Var}(\epsilon_i) = \sigma_{\epsilon_i}^2$;
- Uncorrelated with f : $\text{Cov}(f, \tilde{r}_i) = 0$;
- Uncorrelated with other assets: $\text{Cov}(\tilde{r}_i, \tilde{r}_j) = 0, i \neq j$.

We can show that

$$\text{Var}(\tilde{r}_i) = \underbrace{\beta_i^2 \sigma_f^2}_{\text{systematic risk}} + \underbrace{\sigma_{\epsilon_i}^2}_{\text{unsystematic risk}} .$$